Gamson's Law or a Formateur Premium? Political Parties as Multi-Agent RL Agents Bargaining over Coalition Cabinets

Arpan Jain (B23CS1089) Aradhya Mahajan (B23CS1088)

GitHub: `https://github.com/bot2048/cabinet-bargaining-marl`

1 Question and answer

After an election without a majority, parties bargain over who governs and how the ministries are split. Real coalition cabinets divide portfolios almost exactly in proportion to seats (*Gamson's law* [2]): on 195 real cabinets, share $= 0.031 + 0.896\,s + 0.011\,F$ (s: seat share in the coalition, F: prime minister's party; R^2 0.93). The leading bargaining theory, Baron–Ferejohn (BF) [1], says the *formateur* who proposes the cabinet should take a large premium. **We ask which of the two self-interested parties produce**: we model parties as reinforcement-learning agents rewarded only by their own office and policy payoffs (no fairness term), check them against exact equilibria, vary the institutions, and confront the result with real cabinets. **Answer.** Learning parties favour the formateur whenever the rules give it proposal power, and split by *voting weight* when parties bargain by demands; they never produce seat-proportionality as such. Real cabinets do: no premium relative to seats, and at most a small one (+0.074) relative to voting weights, far below BF.

F1 Proposal bargaining favours the formateur as BF predicts (learner validated against the exact solution), and private values or policy motives only move the premium from the *number* of ministries to their value or to policy control.

F2 Demand bargaining removes the premium but follows voting weights, not seats, exactly as the equilibrium of our protocol, which we solve (pre-registered tests).

F3 Real cabinets follow seats: no detectable premium in counts or key ministries; Gamson's law predicts real splits better than every learned policy.

2 RL formulation

Environment (`cabinet/env.py`, `cabinet/config.py`): a partially observable stochastic game with n parties (seats w_i, majority quota) and m ministries. In a period without a cabinet a formateur $f \sim \rho$ proposes a winning coalition $C \ni f$ and an allocation $\mu : \{1..m\} \to C$; the cabinet forms if every partner accepts and pays per-period flows until the term ends or a member exits. Switchable mechanisms: repeated terms (M1), demand protocol (M2: a recognised first mover, then the others in random order, each demands a share or *closes* a cabinet with the cheapest earlier demands and takes the rest), private types (M4), heterogeneous values (M5), exits (M8), ministerial policy (M9), recognition rule (M10). Level E0 is the exact BF game; E* has every mechanism on.

State $\mathcal{S}$. A *set* of n party tokens (seats, recognition probability, roles, offers, recent cabinets), m ministry tokens (salience, own valuation, policy dimension), pair features, context and decision features; no labels, so one permutation-equivariant network serves any n, m. **Action $\mathcal{A}$** (masked): the formateur includes/excludes each party (masks keep the coalition winning) and points each ministry to a member; voters accept/reject; members stay/exit; under M2 a party demands 0–100% or closes. **Reward $\mathcal{R}$.** $r_i = \sum_{k:\mu(k)=i} \nu_{ik} - \lambda \frac{1}{d} \sum_h |y_h - z_{ih}|$ per governed period (value of own ministries minus policy loss; the holder of a lead ministry sets y_h), an exit bonus on leaving, 0 otherwise; discount δ. Rewards are individual.

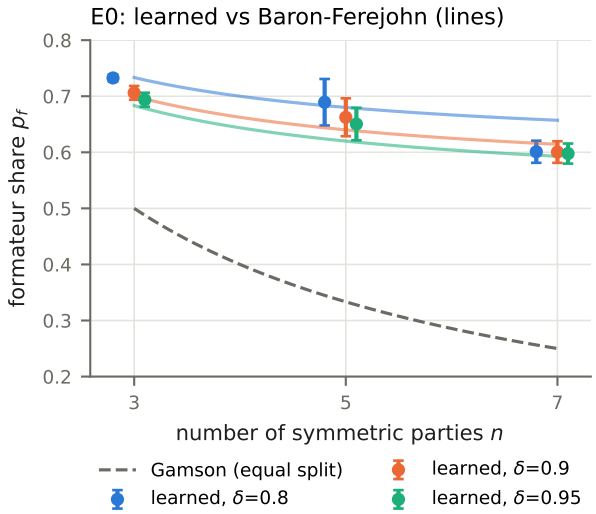


Figure 1: Learner check: learned formateur share (dots) vs the BF closed form (lines) in the symmetric game; dashed: Gamson's equal split.

Algorithm (`cabinet/policy.py`, `cabinet/ppo.py`): one shared attention network with masked autoregressive heads, trained by PPO [11] with GAE on each agent's own decision sequence (semi-MDP); also MAPPO [12], fictitious self-play, PSRO (`cabinet/psro.py`) and a GRU memory. **Baselines**: exact BF (`cabinet/bf_solver.py`), exact demand equilibrium (`cabinet/demand_solver.py`), Gamson, myopic and random agents (`cabinet/baselines.py`). **Metric** (`cabinet/metrics.py`): regression (1) $p = \alpha + \beta s + \gamma F$ on the cabinets formed, γ being the seat-controlled formateur premium, and its analogues for value and policy-ministry shares. 500–2000 PPO iterations per run, 3–10 seeds (the statistical units); hyper-parameters tuned on E0 only.

3 Results

F1, proposal bargaining. *Learner check.* In the symmetric BF game learned shares follow the closed form (Fig. 1; mean absolute gap 0.017, 6/9 cells within 0.02, mean signed gap +0.001), including its decrease with patience δ for $n \le 5$. This matches equilibrium *outcomes*, not equilibrium *play*: learned voters would accept far less than their reservation value δ/n, and against these frozen voters a formateur best response would gain up to 0.153 per formation at $n = 7$ (0.013 at $n = 3$; `scripts/bf_best_response.py`). Formateurs never try low offers, so voters never learn to reject them (a self-confirming pattern [8]); as in laboratory BF experiments [7], proposers get less than theory at $n = 7$. *Institutions.* The base game keeps a premium ($\hat{\gamma} = +0.249$; +0.295 after 4× training). Repetition (even patient, with memory), exits, the recognition rule, MAPPO, fictitious self-play and PSRO leave it in place (Fig. 2). Two

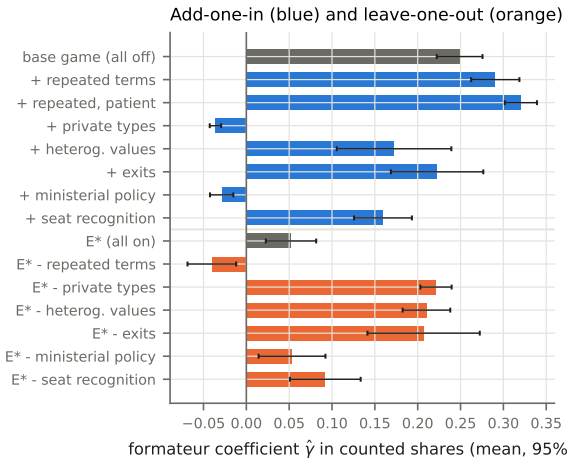


Figure 2: F1: formateur coefficient in counted shares when a mechanism is added to the base game (blue) or removed from E* (orange); mean and 95% CI over seeds.

Table 1: F2: who gets what in fixed parliaments (shares within the cabinet, divisible surplus, $\delta = 0.9$; learned: 3 seeds, all coalitions that form).

	45/35/20 (each pair)	40/30/25/5
Gamson (seats)	largest 0.56–0.69	dummy > 0
demand: exact eq.	0.50/0.50	0.50/0.50, dummy 0
demand: learned	0.50 ± 0.02	0.50 ± 0.03, dummy 0.0%
proposal: weighted BF	form. ≈0.70	form. ≈0.71
proposal: learned	form. 0.65–0.73	form. 0.36–0.71

mechanisms remove the *counted* premium, private types (−0.036) and ministerial policy (−0.029), and removing private types from E* restores it. But the premium moves: in each party's share of the value it places on ministries the formateur coefficient is +0.107 with private types and +0.205 in E*, and with policy it holds the policy-setting ministries (coefficient +0.803). Both persist with 4× training.

F2, demand bargaining and voting weights. On the same random 3–8-party parliaments (10 seeds per protocol, tests fixed in advance in `docs/PREREGISTRATION.md`), demands instead of proposals lower the premium by 0.34 (95% CI 0.32–0.36; exact permutation $p = 5 \times 10^{-6}$) and raise the seat slope by 0.36 (0.34–0.37); the demand premium is within ±0.05 (−0.039, 90% CI −0.047 to −0.030). Is this Gamson's law? Our protocol is not Morelli's [5, 6], so we solved it exactly (backward induction over all demand histories): any two of 45/35/20 seats split **50/50**, because every pair wins, and in 40/30/25/5 the 5-seat party gets nothing; Gamson predicts 0.56/0.44 for 45+35. Learned demand agents reproduce the equilibrium (Table 1): every pair splits within 0.02 of 50/50 (seat slope −0.02 to +0.12) and the dummy is never included. Learned splits follow voting weights; the seat slope of 0.8 in random parliaments arises because seats and voting weights are correlated there. The small negative premium is an exploration artefact (one *close* action faces 101 demand bins): with more exploration of demands it vanishes (−0.009 to +0.003, slope 0.82–0.86; 3 seeds).

F3, real cabinets (Seki–Williams [14] linked to ParlGov [13]; `scripts/seki_eval.py`, `scripts/real_robustness.py`). Relative to seats the prime minister's party has no premium (+0.011, wild-cluster 95% CI −0.023 to +0.045; leave-one-country-out −0.004 to +0.013). Relative to minimal integer voting weights it has one, +0.074 (+0.025 to +0.128), as Ansolabehere et al. report [4]. Seats fit better than weights (R^2 0.93 vs 0.88), and decisively where the two disagree: for 42 pairs of partners with *equal* voting

Table 2: F3: regression (1) with seat shares ($\hat{\beta}, \hat{\gamma}$) and with voting power ($\hat{\gamma}_{bz}$, Banzhaf) on real cabinets and on the cabinets that agents trained only on random parliaments form themselves in the same 195 parliaments (ranges over seeds).

	$\hat{\beta}$	$\hat{\gamma}$ (PM)	$\hat{\gamma}_{bz}$	R^2
real cabinets	0.896	0.011	+0.056	0.93
demand	0.74–0.87	−0.07 to −0.01	−0.08 to −0.02	0.80–0.89
more exploration	0.86–0.87	−0.02 to 0.00	−0.03 to 0.00	0.86–0.87
proposal	0.49–0.55	+0.21–0.34	+0.24–0.38	0.84–0.94
E* (all mech.)	0.63–0.67	+0.01–0.08	+0.05–0.11	0.71–0.73

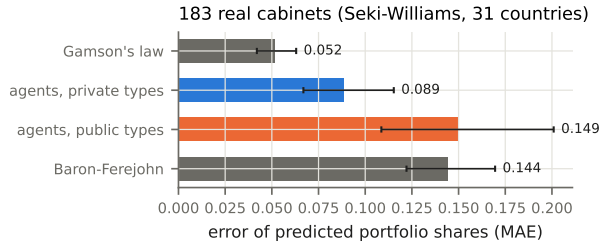


Figure 3: F3: predicting the portfolio split of real coalition cabinets given the real coalition and prime minister; country-clustered 95% CIs.

weight but different seats, share differences follow seat differences with slope 0.84 (95% CI 0.44–1.07; Gamson 1, voting weight 0, our demand agents −0.02 to +0.12), though less clearly without the PM's party (0.52, 24 pairs). Key ministries (finance, foreign, interior, defence, justice; 131 cabinets) show no detectable premium either [3]: 0.589 held vs 0.573 of all ministers (difference +0.015, 95% CI −0.021 to +0.056). Given the real coalition and prime minister, Gamson's law errs by 0.052 per party, BF by 0.144 and agents with private types by 0.087–0.090, every one worse than Gamson (Fig. 3). Left to bargain freely in the real parliaments (Table 2), demand agents with more exploration reproduce the real seat regression almost exactly, although their rule is voting weight (F2; in most real parliaments the two are close); E* matches the small premium relative to voting power; proposal agents overshoot.

Reliability. Best responses trained by PPO (one seat learns against the frozen profile) gain +0.015 per period against E* (10% of a seat's payoff) and 19% and 18% against demand and proposal agents on random parliaments, +0.037 and +0.088 against scripted Gamson and BF-heuristic profiles. In the BF game they gain 0.040 (20%), more than the analytic formateur deviation, so the test detects the gaps found above: learned profiles are approximate, not exact, equilibria. Cross-play between independent runs costs +0.0020 per period.

4 Implications and limitations

To the question of the project: self-interested learning parties do not follow seats as such. They favour the formateur as far as the rules give it proposal power, and otherwise split by voting weight; information and policy motives change the currency of the premium, not its existence. Demand bargaining comes closest to real cabinets, but these follow seats even where voting weights disagree, so Gamson's law likely also rests on rules outside these models, such as commitments before the split [9] or within-party bargaining [10]. **Limitations**: scaled-down sizes; approximate equilibria; the PM's party proxies the formateur; real shares count ministers (salience only via key ministries). **Not done**: party-specific manifesto salience (needs registered data access), a two-stage close/demand head (retraining), more seeds (every pre-registered effect is decisive at 10). Claims and sources: `docs/AUDIT.md`, `docs/SPEC_COVERAGE.md`.

References

[1] D. Baron, J. Ferejohn. Bargaining in legislatures. *APSR* 83(4), 1989.

[2] W. Gamson. A theory of coalition formation. *Am. Sociol. Rev.* 26(3), 1961.

[3] P. Warwick, J. Druckman. The portfolio allocation paradox. *EJPR* 45(4), 2006.

[4] S. Ansolabehere, J. Snyder, A. Strauss, M. Ting. Voting weights and formateur advantages in the formation of coalition governments. *AJPS* 49(3), 2005.

[5] M. Morelli. Demand competition and policy compromise in legislative bargaining. *APSR* 93(4), 1999.

[6] M. Montero, J. Vidal-Puga. Demand commitment in legislative bargaining. *APSR* 101(4), 2007.

[7] G. Fréchette, J. Kagel, M. Morelli. Gamson's law versus non-cooperative bargaining theory. *Games Econ. Behav.* 51(2), 2005.

[8] D. Fudenberg, D. Levine. Self-confirming equilibrium. *Econometrica* 61(3), 1993.

[9] A. Kamm, S. Siegenthaler. Commitment timing in coalitional bargaining. *Exp. Econ.* 27, 2024.

[10] G. Cox. Nonunitary parties, government formation, and Gamson's law. *APSR* 115(3), 2021.

[11] J. Schulman et al. Proximal policy optimization algorithms. arXiv:1707.06347, 2017.

[12] C. Yu et al. The surprising effectiveness of PPO in cooperative multi-agent games. NeurIPS, 2022.

[13] H. Döring, P. Manow. ParlGov: Parliaments and governments database. parlgov.org.

[14] K. Seki, L. Williams. Updating the party government data set. *Electoral Studies* 34, 2014.